

Naive Bayes Algorithms

A COMPREHENSIVE & BASIC GUIDE

Understanding Probability in Machine Learning

What is Naive Bayes?

A Classification Technique

Naive Bayes is a collection of classification algorithms based on **Bayes' Theorem**.

It is not a single algorithm but a family of algorithms where all of them share a common principle.

Why "Naive"?

It is called "Naive" because it makes a **simple assumption** that every pair of features being classified is independent of each other.

Probabilistic Classifiers

Unlike some algorithms that draw a strict line between groups (like Support Vector Machines), Naive Bayes is **probabilistic**.

It calculates the probability that a given input belongs to a certain category.

Example: "There is a 92% chance this email is Spam and an 8% chance it is Not Spam."



Predicts

Likelihood of
outcomes



Categorizes

Based on highest
probability

Real World Analogy

Imagine you are sorting fruits based on features.

If an object is **Yellow**, **Long**, and **Sweet**, you would classify it as a **Banana**.

Naive Bayes calculates the probability of it being a Banana given these three specific features separately.



The Origin



Reverend Thomas Bayes

(1702–1761)

An English statistician and philosopher who formulated the specific case of the theorem that bears his name.

His work on the probability of causes was published posthumously, laying the groundwork for modern statistical inference.

Probability Basics



$P(A)$

Prior

Probability

The probability of an event occurring before new data is collected.

Ex: Probability of rolling a 6 is 1/6.



$P(B)$

Evidenc

e

The total probability of the data (evidence) occurring.

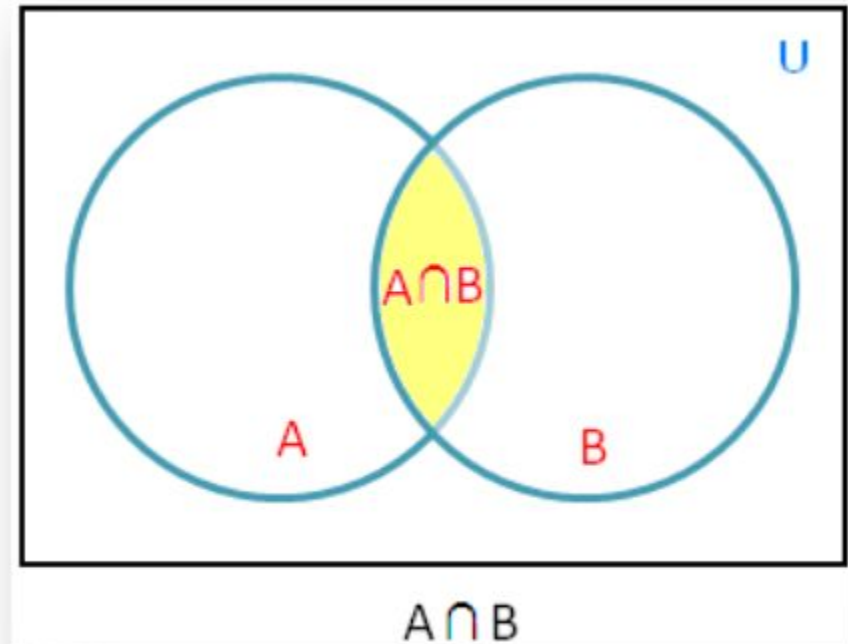
Conditional Probability

Definition

The likelihood of an event or outcome occurring, based on the occurrence of a previous event or outcome.

$$P(A | B)$$

Read as: "The probability of A given B".



Example: Drawing Cards

- ✓ **Scenario:** You draw a card from a standard deck.
- ✓ **Event A:** The card is a Queen.
- ✓ **Event B:** The card is a Face Card (J, Q, K).
- ✓ **Question:** What is the probability it is a Queen, *given that* we know it is a Face Card?
- ✓ Since there are 12 face cards and 4 of them are Queens:

$$PQ \mid \text{Face} = \frac{4}{12} = \frac{1}{3}$$

Bayes' Theorem

The mathematical formula for calculating conditional probability.

$$P(A | B) = \frac{P(B | A) \times P(A)}{P(B)}$$

This is the foundation of the Naive Bayes Classifier.

1. Posterior Probability

$$P(A | B)$$

The Goal

This is what we are trying to calculate.

It is the probability of the hypothesis (A) being true given the observed evidence (B).

"Probability that this email is Spam (A) given that it contains the word 'Free' (B)."

2. Likelihood

$$P(B | A)$$

The Evidence Strength

The probability of seeing the evidence (B) if the hypothesis (A) is true.

"If an email IS Spam (A), what is the probability it contains the word 'Free' (B)?"

3. Prior Probability

PA

Initial Belief

The probability of the hypothesis being true regardless of the data.

"What is the overall probability that ANY email is Spam?"

(Usually calculated from the training data frequency).

4. Marginal Likelihood

P_B

The Evidence Probability

The probability of observing the evidence (B) regardless of the hypothesis.

In classification, this is often a constant denominator for all classes, so we can sometimes ignore it during comparison.

Why "Naive"?



Simple

It makes a very strong, simplistic assumption.



Independent

It assumes all features are unrelated.

Assumption of Independence

Naive Bayes assumes that the presence of a particular feature in a class is **unrelated** to the presence of any other feature.

Reality

In an apple, color (Red) and shape (Round) might be related.

Naive View

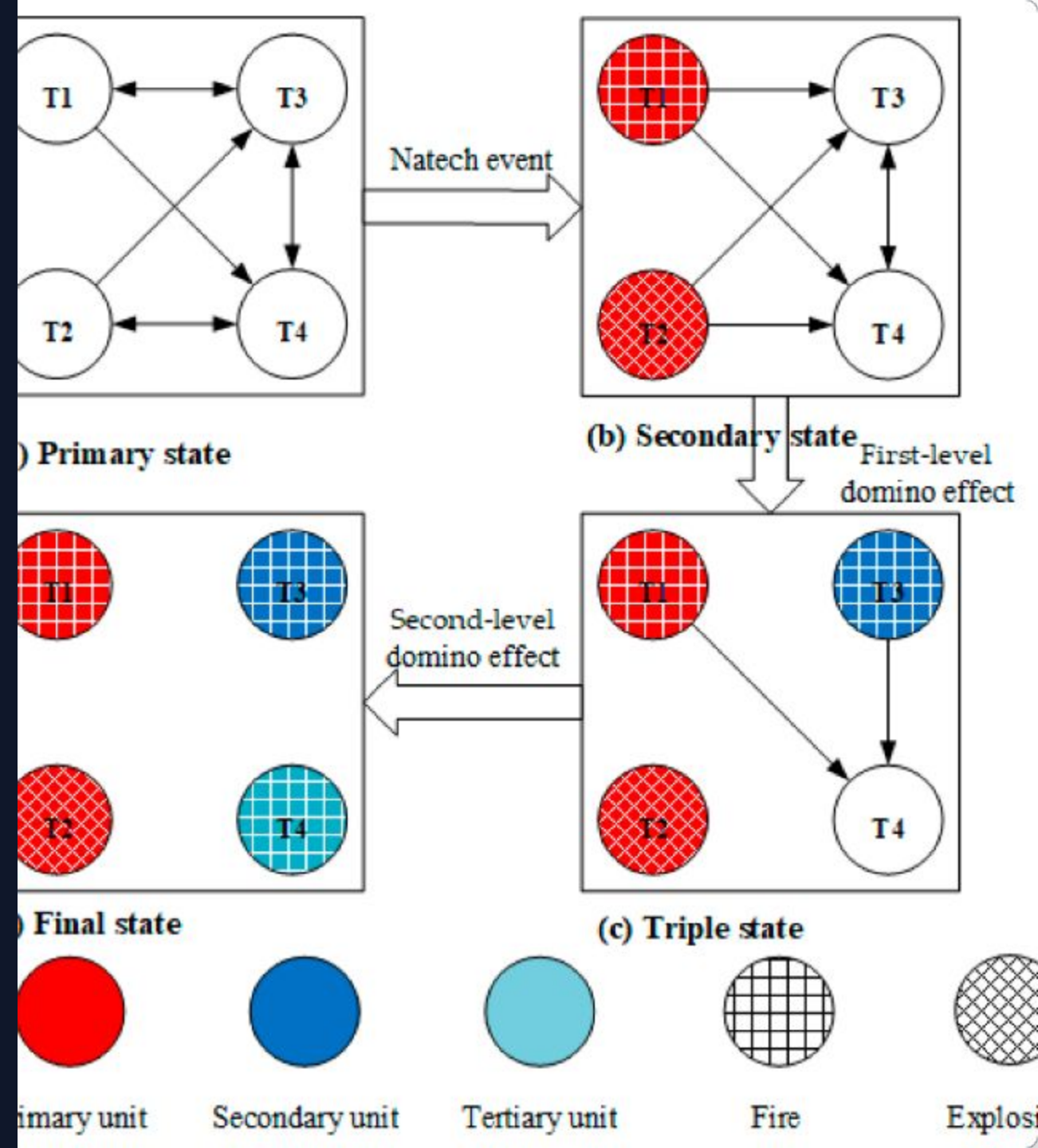
The algorithm considers "Red" and "Round" as two completely separate contributors to the probability of it being an apple.

Visualizing Independence

Think of independent events like dominoes that DON'T touch.

One falling does not cause the other to fall. They are treated as separate isolated events.

This simplifies the math drastically.



Step 1: The Dataset

We start with a dataset containing features and a target class.

Weather (Feature)	Play Tennis (Target)
Sunny	No
Overcast	Yes
Rainy	Yes
Sunny	Yes

Step 2: Frequency Tables

We convert the dataset into a frequency table.

Play Counts

- ✓ Yes: 9
- ✓ No: 5
- ✓ Total: 14

Weather Counts

- ✓ Sunny: 2 Yes, 3 No
- ✓ Overcast: 4 Yes, 0 No
- ✓ Rainy: 3 Yes, 2 No

Step 3: Likelihood Table

We calculate the probabilities for each feature.

Weather	No	Yes	P(Weather)
Sunny	3/5 (0.6)	2/9 (0.22)	5/14 (0.36)
Overcast	0/5 (0.0)	4/9 (0.44)	4/14 (0.29)
Rainy	2/5 (0.4)	3/9 (0.33)	5/14 (0.36)

Step 4: Calculation

Using the tables, we apply the Naive Bayes equation.

For a new instance (e.g., Weather = Sunny), we calculate:

$$P (\text{Yes} | \text{Sunny}) \propto P (\text{Sunny} | \text{Yes}) \times P (\text{Yes})$$

We compare the result for "Yes" vs "No" and pick the highest.

3 Major Types



Gaussian

Continuous
Data



Multinomial

Count
Data



Bernoulli

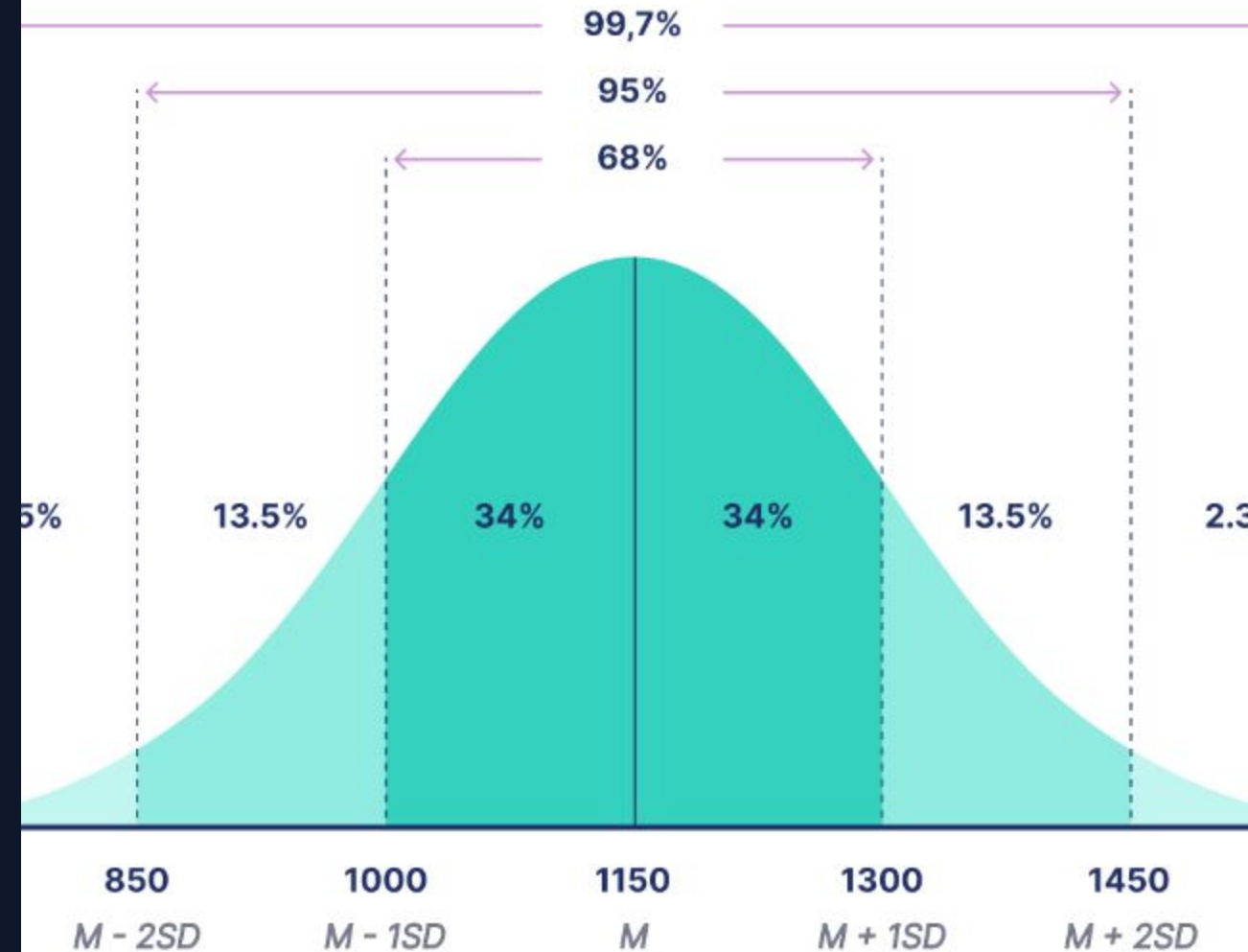
Binary
Data

Gaussian Naïve Bayes

Used when features are continuous values (like height, weight, foot size) and follow a normal distribution.

We assume the data forms a Bell Curve.

Using the empirical rule in a normal distribution



Gaussian Use Cases

- ✓ **Iris Flower Classification:** Predicting flower species based on petal width (cm) and length (cm).
- ✓ **Medical Diagnosis:** Classifying health risks based on blood pressure (numeric) and cholesterol levels.

Multinomial Naive Bayes

Used for discrete counts.

It is the "Go-To" algorithm for **Text Classification**
(Natural Language Processing).

It counts how many times a word appears in a document.

Multinomial Use Cases

Topic Categorization

Sorting news articles into "Sports", "Politics", or "Tech" based on word frequency.

Sentiment Analysis

Determining if a movie review is "Positive" or "Negative" based on word choices.

Bernoulli Naive Bayes

Similar to Multinomial, but it only cares about the **presence** or **absence** of a feature (0 or 1).

It doesn't count *how many times* a word appears, just *if* it appears.

Bernoulli Use Cases

- ✓ **Short Text Classification:** Analyzing headlines or tweets where words rarely repeat.
- ✓ **Keyword Triggering:** Detecting if specific banned words exist in a comment (Yes/No).

Example 1: Spam Filtering

The most famous application of Naive Bayes.

We want to classify an email as **SPAM** or **HAM** (Not Spam).



"Bag of Words"

The algorithm treats an email as a bag of mixed-up words. It ignores grammar and order.

Email 1

"Win cash prize now"

Features

{Win, Cash, Prize, Now}

The Calculation

We calculate:

$$P_{\text{Spam}} | \text{Words} \propto P_{\text{Spam}} \times \prod P_{w_i} | \text{Spam}$$

We multiply the probability of finding each word (w) in a Spam email.

The "Zero Frequency" Problem

What if a word never appeared in training?

If $P(\text{Word}|\text{Spam}) = 0$, the whole multiplication becomes 0.

The Fix: Laplace Smoothing

We add 1 to every count so no probability is ever zero. This is a small but crucial adjustment.

Example 2: Play Tennis?

Predicting if we should play tennis based on the weather.

Features: Outlook, Temperature, Humidity, Wind.



The Question

Today's Conditions

- ✓ Outlook: Sunny
- ✓ Temp: Cool
- ✓ Humidity: High
- ✓ Wind: Strong

Prediction?

Yes or

No

Step A: Priors

From our dataset of 14 days:

✓ Days we played (Yes): 9

✓ Days we didn't (No): 5

$$P(\text{Yes}) = 9/14 = 0.64$$

$$P(\text{No}) = 5/14 = 0.36$$

Step B: Likelihoods for "Yes"

Probability of conditions given we played:

- ✓ $P(\text{Sunny} \mid \text{Yes}) = 2/9$
- ✓ $P(\text{Cool} \mid \text{Yes}) = 3/9$
- ✓ $P(\text{High Humidity} \mid \text{Yes}) = 3/9$
- ✓ $P(\text{Strong Wind} \mid \text{Yes}) = 3/9$

Step C: Likelihoods for "No"

Probability of conditions given we DID NOT play:

- ✓ $P(\text{Sunny} \mid \text{No}) = 3/5$
- ✓ $P(\text{Cool} \mid \text{No}) = 1/5$
- ✓ $P(\text{High Humidity} \mid \text{No}) = 4/5$
- ✓ $P(\text{Strong Wind} \mid \text{No}) = 3/5$

Calculate P(Yes | Conditions)

$P(\text{Yes}) * P(\text{Sunny}|\text{Yes}) * P(\text{Cool}|\text{Yes}) * P(\text{High}|\text{Yes}) * P(\text{Strong}|\text{Yes})$

$$0.64 \times 0.22 \times 0.33 \times 0.33 \times 0.33 = 0.0053$$

Calculate P(No | Conditions)

$P(\text{No}) * P(\text{Sunny}|\text{No}) * P(\text{Cool}|\text{No}) * P(\text{High}|\text{No}) * P(\text{Strong}|\text{No})$

$$0.36 \times 0.6 \times 0.2 \times 0.8 \times 0.6 = 0.0206$$

Since $0.0206 > 0.0053$, we predict **NO**.

Advantages

- ✓ **Super Fast:** Great for real-time predictions.
- ✓ **Less Data Needed:** Works well even with smaller training sets compared to other algorithms.
- ✓ **Simple:** Easy to implement and understand.



Conclusion

Naive Bayes is a foundational algorithm in Machine Learning.

Despite its "Naive" assumption of independence, it performs remarkably well for complex tasks like Spam Filtering and Sentiment Analysis.

It remains a key tool in the Data Science toolkit.

